MPFI:

In this system for injecting fuel we need to measure the amount of air inside the system. Electronic injection system consists of various devices. Air flow meter (sensor) is placed in the system for measuring the amount of air inducted and the signal is sent to the ECU. Then this ECU sends one signal to the injectors. By the method injectors injects fuel in uniform manner with the help of pump in to the each cylinder separately respective to the signal from the ECU.

Types of MPFI System:

Based on the amount of air inducted in to system MPFI classified in to two types

1. D-MPFI (manifold pressure control type)
2. L-MPFI (air flow control type)

D-MPFI

D-MPFI also called as a D-Jetronic. D-Jetronic is a german word comes from a word 'Druck' (pressure). In the type of system air flow rate is measure based on the Vacuum created inside the manifold. D-MPFI is mainly used in the engine which was controlled by the computer control system.

L-MPFI

L-MPPFI also called as L-Jetronics. The word L in the L-Jetronics is comes from the german word called 'Luft'. The meaning for 'Luft' is air. This L-MPFI system is mainly used in the electronic fuel injection (EFI) and Computer controlled system (CCS) adopted engines. In this type of system the air flow rate inside the manifold is measured using sensor called Air flow meter.

Basic parts of MPFI:

The MPFI are consists of three different parts:

1) Electronic Control System
2) Fuel System
3) Air Induction System

(i) MPFI – Electronic Control System

It consists various sensors like Air flow meters, Water temperature sensor, throttle position sensor, Intake Air Temperature sensors etc. The ECU system controls the injection duration of the injectors based on the signals received from the above sensors.

(ii) MPFI – Fuel System

It consists of Fuel Pump, fuel cold start injector, Timing and Injection signal control (ECU). The ECU system controls and defines the amount of fuel to be injected in to the manifold with the help of injector by receiving signals from the sensors.

(iii) MPFI – Air Induction System

It consists of Air cleaner, Air flow meter, Throttle body, Air valve, Air intake chamber, Intake manifold and cylinder. All this devices are used in the system is for supplying exact amount of air required for the complete combustion.

9.8 METHOD OF FUEL SUPPLY (DIESEL ENGINE)

In the present economic condition spending more money for buying fuel for the vehicles becomes a difficult one for the

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